OSAMA BIN HABIB

+1 571 547-7279 - habib_osama@yahoo.com - LinkedIn - Portfolio - Sterling, VA

EXPERIENCE

AI Solutions Architect & Implementation Engineer
Mar 2026 – Present

North American Management · Alexandria, VA · Hybrid

- Own end-to-end implementation of AI-powered platform for genealogical research, building custom integrations and automating ingestion of 1,000+ historical records — reducing insight retrieval time by 40%.
- Design intelligent workflows using LLM orchestration and RAG achieving 90%+ accuracy and 99.9% uptime in production.
- Implement RBAC, audit logging, PII handling, and encryption ensuring 100% data integrity and zero compliance findings.
- Build tooling and automation, increasing data throughput by 45% and reducing ad-hoc analytics time by 40%.
- Built interactive admin and analytics dashboards in **React**, enabling non-technical stakeholders to query the knowledge base, review audit logs, and manage access permissions through a unified UI.
- Built evaluation frameworks for production LLMs, improving LLM accuracy by 25% and establishing guardrails for safe customer deployment.
- Built a LangGraph agentic research workflow that classifies questions, retrieves genealogical records, generates and verifies answers, and supports human approval before delivery, improving answer accuracy by 70%.
- Used GitHub Copilot and Codex to assist with implementation, code exploration, and iterative development of the platform, automating 60% of manual workflows and iterations.

Software Engineer II (Forward Deployed) | AI/ML | Enterprise Clients
Aug 2022 – Apr 2025

Afiniti · Fortune 500 Clients: UnitedHealth Group, DirecTV, American Express -Arlington, VA

- Technical lead for customer deployments of AI automation platform across Fortune 500 accounts (UHG, DirecTV, AmEx) — owned scoping, integrations, and go-live, reducing manual processes by 8–15% at 99.9% uptime.
- Designed custom integrations with enterprise systems (EHRs, CRMs, phone systems) using Python, TypeScript, and REST APIs, enabling real-time patient and customer interaction workflows.
- Created reusable implementation playbooks and accelerators, cutting time-to-value by 25–30% across 5+ global accounts and increasing platform adoption by 20%+.
- Debugged complex integration issues across application, data, and infrastructure layers, reducing post-launch escalations by 35–40%.
- Partnered with Sales and Product to scope technical requirements, deliver demos, and influence roadmap based on field learnings.

Associate Software Engineer (Forward Deployed) | ML Pipeline Delivery
Apr 2021 – Aug 2022

Afiniti

- Built automated ETL pipelines and real-time APIs reducing manual effort by 60% and accelerating data movement by 40%; established CI/CD and code review standards across the team.

TECHNICAL PROJECTS & VENTURES

PriorAuth AI Assistant — Clinical Decision Support Tool

- Production-grade LLM with RAG and agentic workflow cutting manual review by 95% and flagging 100% of policy mismatches; human-in-the-loop audit trails for compliance. Delivered end-to-end: APIs, prompt engineering, and knowledge transfer to client teams. (App Link)

Disease Outbreak Prediction Platform — Best Track Award, AI Expo 2025

- Built AEGIS — a disease outbreak prediction app using Claude to analyze multi-source data (Reddit, CDC, Delphi, Google Trends) detecting early signals of flu, measles, and COVID; delivered real-time alerts enabling public health officials to make faster, data-driven decisions with 85% prediction accuracy. **Won Best Track Award at AI Expo 2025 Hackathon.** (**App Link**)

OHOK — Co-Founder & Technical Lead
2021 – 2022

- Founded a software company building production AI and full-stack products for clients — spanning custom integrations, intelligent workflow automation, and scalable backend systems; directly informed client-facing engineering instincts applied across all roles above.

SKILLS

Forward Deployed Engineering: Customer Implementations, Technical Scoping, Go-live, Integration, Intelligent Workflows, Debugging, Playbooks, Tooling & Automation

Healthcare & Compliance: EHR/PMS Integrations, HIPAA, PHI/PII, Audit Logging, Clinical Workflows

Technical: Python, TypeScript, SQL, REST APIs, Microservices, LLM/RAG, Vector DBs, Docker, Kubernetes, GCP, AWS, CI/CD, React

AI Tools: Claude Code, Codex, GitHub Copilot, GPT 6 Luna, DeepSeek, LangGraph, Agentic AI

EDUCATION & CERTIFICATIONS

MS Data Analytics: McDaniel College · 2027 · GPA 3.83/4.0

MS Business Analytics: Georgetown University · 2024 · GPA 3.52/4.0

BS Computer Science: Institute of Business Administration, Karachi · 2020

Certifications: IBM Python & SQL · MIT Data Science · AWS Academy Graduate